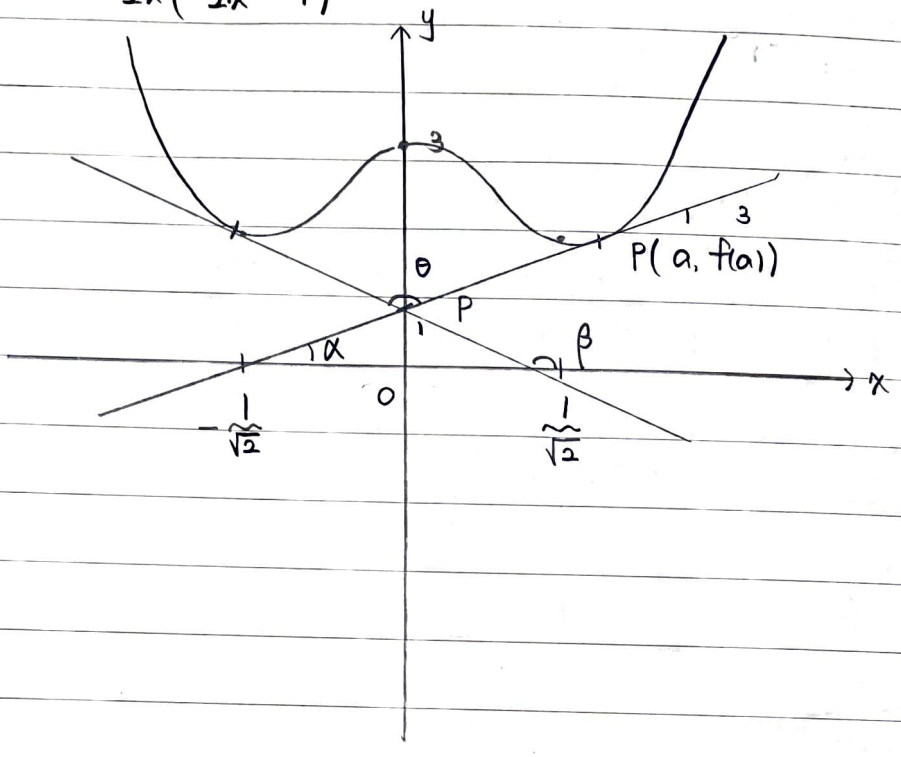
10. 접선 : $\frac{4}{3}(x+1)+4-y=0$

법선 : $-\frac{3}{4}(x+1)+4-y=0$

11. $f(x) = x^4 - x^2 + 3$

$f'(x) = 4x^3 - 2x$

$= 2x(2x^2-1)$



x	$\cdots$	$-\frac{1}{\sqrt{2}}$	$\cdots$	0	$\cdots$	$\frac{1}{\sqrt{2}}$	$\cdots$
$f'(x)$	$-$	0	$+$	0	$-$	0	$+$
$f(x)$	↘	$3-\frac{1}{4}$	↗	3	↘	$3-\frac{1}{4}$	↗

$\frac{f(a)-1}{a-0} = f'(a)$

$\frac{a^4-a^2+2}{a} = 4a^3 - 2a$

$3a^4 - a^2 - 2 = 0$

$a = \pm 1$

$\therefore$ $P(1, 3)$ or $P(-1, 3)$

$\tan\alpha = 2$, $\tan\beta = -2$

$\tan\theta = \tan(\beta-\alpha) = \frac{-4}{1-4} = \frac{4}{3}$ //